R16

Code No: 135CH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, November/December - 2018
FUNDAMENTALS OF MECHANICAL ENGINEERING
(Common to CE, EEE, CSE, MIE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART - A

(25 Marks)

1.a) Why work is called high grade energy and heat is called a low-grade energy? [2]
b) What are the limitations of first law of thermodynamics? [3]
c) Define Charle's law. [2]
d) What are saturation states? [3]
e) Define Carnot cycle. [2]
f) What is the compression ignition engine? Why is the compression ratio of such an engine more than that of SI engine? [3]
g) What is priming. [2]
h) What is a difference between Air-conditioning and refrigeration? [3]
i) Define strength and stiffness. [2]
j) What is an open belt drive? Write its applications. [3]

PART - B

(50 Marks)

2.a) A stationary mass of gas is compressed without friction from an initial state of $0.3m^3$ and $0.105m^3$ MPa to a final state of $0.15m^3$ and $0.105m^3$ MPa, the pressure remaining constant during the process. There is a transfer of 37.6 kJ of heat from the gas during the process. How much does the internal energy of the gas change?
b) What are the factors which influence ozone depletion? [5+5]

OR

3.a) Define enthalpy, why does the enthalpy of an ideal gas depends only on temperature.
b) Write a sketch explain the working of Solar photovoltaic cell. [5+5]

4.a) Water at 40^0C is continuously sprayed into a pipeline carrying 5 tonnes of steam at 5 bar, 300^0C per hour. At a section, downstream where the pressure is 3 bar, the quality is to be 95%. Find the rate of water spray in kg/h.
b) What is feed pump, super heater and Economizer? [5+5]

OR

5.a) Derive the expression for the difference in heat capacity, Cp and Cv.
b) Explain the working of Locomotive Boiler. [5+5]

6.a) A heat engine absorbs 100J of heat from the hot reservoir and releases 60 J of heat to the cold reservoir during each cycle.
i) What is its efficiency?
ii) If each cycle takes 0.50s, find the power output of this engine.
b) What is the difference between CI engine and SI engine? [5+5]

OR

7.a) What is a heat engine how it is Classified.
b) A Diesel engine has compression ratio of 14 and cut-off takes place at 6% of the stoke. Find the air standard efficiency. [5+5]

8.a) With a neat sketch explain the operation of Reciprocating pump.
b) Explain about vapour compression refrigeration system with neat sketch. [5+5]

OR

9.a) What is the multi-stage compression? What are the advantages of multistage over single stage compression?
b) Define dew point temperature, Absolute humidity, wet bulb depression. [5+5]

10.a) Define clutch and shaft. List the advantages and applications.
b) With a neat sketch explain the working of any one Absorption dynamometers. [5+5]

OR

11.a) Write the applications of silica, ceramics, glass and diamond.
b) What is shoe brake and band brake list out the differences? [5+5]

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